

# SIH 2026 — PS 26168: AI-ML Intelligent Dead Reckoning

Project brief & selection playbook — Jatin Sharma

## Why ISRO/SAC published this problem statement

- **NavIC is currently degraded.** IRNSS-1F's atomic clock failed (Mar 2026) after reaching design life; as of mid-2026 NavIC has been down to 3 functional satellites — below the 4 needed for standalone fix. A resilient, sensor-fusion fallback that isn't dependent on one constellation is now more urgent, not less.
- **GNSS jamming/spoofing in India is measured and rising.** DGCA/AAI logged 1,951 GNSS RFI reports (Nov 2023–Nov 2025), an increasing trend; separately, 465+ jamming/spoofing incidents were recorded along the India-Pakistan border (late 2023–early 2025), affecting Amritsar, Jammu, even Delhi airspace.
- **The fleet-reality gap.** High-end cars have factory-fitted, wheel-connected INS. Trucks, older cars, and millions of two-wheelers — the vast majority of Indian vehicles — have none, and rely solely on a dashboard-mounted phone. ISRO wants a software-only, phone-sensor-only solution because that's the only thing deployable across that fleet without hardware retrofits.

## What SAC-ISRO mentors are likely evaluating

Mentors are from ISRO's PNT/navigation applications group, not a consumer-product team. Prioritize technical rigor and quantified results over UI polish.

**Explicit screening requirement (from the PS):** teams must submit preliminary AI models **and a position plot inferred from an IO-VNBD subset** as part of the proposal. Most teams skip this and only submit slides — a working plot (even rough) is the single highest-leverage artifact you can produce.

## Recommended architecture (scoped for ~1 month build)

| Module                   | Approach                                                                       | Why                                                                             |
|--------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| In-vehicle alignment     | Gravity/heading-based auto-calibration at each ZUPT stop                       | Handles arbitrary mount angle without manual setup                              |
| Speed / vibration filter | Lightweight CNN or CNN+BiLSTM on windowed accel/gyro (10Hz)                    | Matches Cortés et al.'s smartphone strapdown speed-net (<13k params, real-time) |
| Heading fusion           | Gyro + compass, classical filter                                               | Cheap, reliable, no training data needed                                        |
| Map-matching             | UKF + Hidden Markov map matching onto OpenStreetMap; non-holonomic constraints | Directly requested in the PS; snaps drift back to road grid                     |
| GNSS+INS fusion          | Kalman filter backbone + learned correction / noise-adaptation layer           | Mirrors AI-IMU Dead-Reckoning (1.10% translational error on KITTI, IMU-only)    |

## Priority order to get selected (internal round, this month)

1. **Lead with incident data.** Open the deck with the NavIC shortfall + 1,951 RFI reports + 465 border incidents. Almost no other team will have sourced this — it signals real domain research.
2. **Produce the required position-drift plot first.** Train even a rough speed model on one IO-VNBD sequence, integrate to a trajectory, plot estimated vs ground-truth path, report drift %. This is explicitly asked for in the PS and most teams will skip it.
3. **Cite validated prior art for your design choices.** Reference Cortés/Solin/Kannala's CNN speed-net and the AI-IMU Dead-Reckoning paper's KF + learned-noise approach — shows the architecture is grounded, not guesswork, scoped realistically for a month.
4. **Demonstrate the GNSS↔INS handoff explicitly.** The PS stresses a 'seamless' transition; show mode-switch latency and smoothing behavior, not just steady-state drift accuracy.
5. **Keep the deck tight.** ~5-6 slides, 5 min + viva is typical for internal rounds. One architecture diagram, one drift plot, one incident-data slide, clear narrative — no filler.

## Key references

- Cortés, Solin, Kannala — "Deep Learning Based Speed Estimation for Constraining Strapdown Inertial Navigation on Smartphones," MLSP 2018 (arXiv:1808.03485)
- Brossard et al. — "AI-IMU Dead-Reckoning" — Kalman filter + DNN noise adaptation, 1.10% translational error on KITTI using IMU only
- Xiao, Ren, Li — "An Inertial Sequence Learning Framework for Vehicle Speed Estimation via Smartphone IMU" (DVSE), arXiv:2505.18490
- IO-VNBD dataset: [github.com/onyekpeu/IO-VNBD](https://github.com/onyekpeu/IO-VNBD)